WORKED SOLUTION

STEP II 2017, Question 9

Mechanics • Friction and Rigid Bodies

STEP II 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q9

The key idea

Each cylinder is a rigid body acted on by forces at only two places, the plank and the floor, so taking moments about its centre is the natural first move: the two normal reactions pass through the centre, which leaves only the two frictional forces, and they must be equal. After that it is a matter of resolving for the cylinder and for the plank, and then feeding in the friction laws $G \leq \frac{1}{2}R$ and $F \leq \frac{1}{2}N$ at the contacts. The geometry links θ to a and r at the very end.

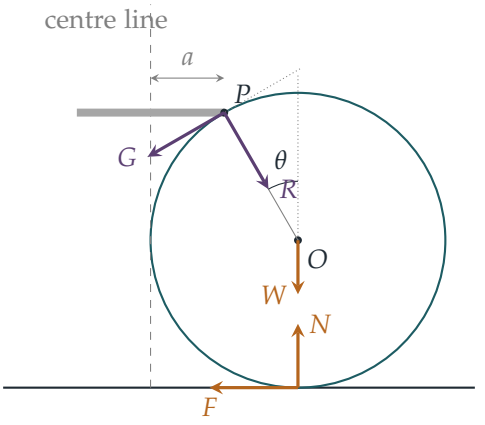
Solution

Model and set-up. Treat the cross-section as two discs of radius r and a horizontal rod, with all contacts rough. The cylinders are a negligible distance apart, so their centres are at horizontal distance r either side of the vertical line of symmetry. The plank has width $2a < 2r$, so it rests with its two long edges on the inner upper parts of the cylinders. By symmetry we need only consider the right-hand cylinder.

Let P be the point where the plank touches it. The tangent at P makes angle θ with the horizontal plank, so the radius to P makes angle θ with the vertical. The forces on the cylinder are:

- its weight W, acting at the centre O;
- at the floor, the normal reaction N upwards and friction F (acting towards the centre line, since the plank pushes the cylinder outwards);
- at P, the normal reaction R along PO, and a frictional force G along the tangent, directed downwards and inwards.

By Newton's third law, the plank feels R along OP (upwards, away from the centre line) and G along the tangent, upwards and outwards.



The dotted line through P is the tangent. Horizontally, P is $r\sin\theta$ inside the centre O, and O is at distance r from the centre line.

Part (i)

Moments about O for the cylinder. W, N and R all pass through O. Both frictional forces are tangential to the circle, so each has moment (force)$\times r$, and they turn in opposite senses: F at the bottom, pointing inwards, and G at P, pointing down and inwards. Hence

$$Fr = Gr \quad \Longrightarrow \quad G = F.$$

Resolving horizontally for the cylinder (outwards positive): R has horizontal component $R\sin\theta$ outwards, G has component $G\cos\theta$ inwards and the floor friction F acts inwards, so

$$R\sin\theta - G\cos\theta - F = 0.$$

With $G = F$ this is

$$R\sin\theta = F(1+\cos\theta),$$

as required.

The deduction. There is no slipping at the plank, so friction there satisfies $G \le \frac{1}{2}R$, that is $F \le \frac{1}{2}R$. Then

$$R\sin\theta = F(1+\cos\theta) \le \tfrac{1}{2}R(1+\cos\theta),$$

and dividing by $\frac{1}{2}R > 0$ (the plank presses on the cylinder) gives $2\sin\theta \le 1+\cos\theta$.

Part (ii)

Resolving vertically for the plank. Each edge receives R, with upward component $R\cos\theta$, and $G = F$ along the tangent upwards, with upward component $F\sin\theta$. So

$$2(R\cos\theta + F\sin\theta) = kW.$$

Using part (i), $R = \dfrac{F(1+\cos\theta)}{\sin\theta}$, so

$$R\cos\theta + F\sin\theta = \frac{F\left[(1+\cos\theta)\cos\theta + \sin^2\theta\right]}{\sin\theta} = \frac{F(1+\cos\theta)}{\sin\theta},$$

since $\cos^2\theta + \sin^2\theta = 1$. Hence

$$\tfrac{1}{2}kW = \frac{(1+\cos\theta)}{\sin\theta}\,F.$$

(The plank's horizontal forces cancel by symmetry, as do its moments.)

Resolving vertically for the cylinder. The reactions from the plank push down by $R\cos\theta + G\sin\theta = \frac{1}{2}kW$, so

$$N = W + \tfrac{1}{2}kW = \left(\frac{2}{k}+1\right)\tfrac{1}{2}kW = \left(1+\frac{2}{k}\right)\left(\frac{1+\cos\theta}{\sin\theta}\right)F.$$

That is the required expression. (This is just the obvious fact that the floor carries half of the total weight $2W + kW$.)

No slipping on the floor. The condition is

$$F \le \tfrac{1}{2}N.$$

From the expression for N,

$$\frac{N}{F} = \left(1+\frac{2}{k}\right)\frac{1+\cos\theta}{\sin\theta}.$$

Part (i) gives $\dfrac{1+\cos\theta}{\sin\theta} \ge 2$, because $\sin\theta > 0$ for acute θ, and $1 + \frac{2}{k} > 1$ because $k > 0$. So $N/F > 2$, i.e. $F < \frac{1}{2}N$. The floor condition therefore holds automatically once the plank does not slip, so it puts no extra restriction on θ.

Ask yourself Why does the floor contact come for free, when both contacts carry the same friction F?

Because the floor presses harder. Comparing the formulas for R and N gives $N = \left(1+\frac{2}{k}\right)R > R$, since the floor also carries the cylinder's own weight. With the same friction at both contacts and a larger normal force at the floor, the plank contact is always the one that slips first.

Part (iii)

Rearrange the inequality from part (i) and square. Both sides are positive, so squaring keeps the direction:

$$4\sin^2\theta \le (1+\cos\theta)^2 \iff 4(1-\cos\theta)(1+\cos\theta) \le (1+\cos\theta)^2.$$

Dividing by $1+\cos\theta > 0$ gives $4 - 4\cos\theta \le 1 + \cos\theta$, i.e. $\cos\theta \ge \frac{3}{5}$. Since θ is acute,

$$\sin\theta = \sqrt{1-\cos^2\theta} \le \sqrt{1-\tfrac{9}{25}} = \tfrac{4}{5}.$$

(Equivalently, with $t = \tan\frac{1}{2}\theta$ the inequality reads $2t \le 1$, and $\sin\theta = \frac{2t}{1+t^2}$ is at most $\frac{4}{5}$ when $t = \frac{1}{2}$.)

The geometry. The edge P of the plank is at horizontal distance a from the centre line. The centre O is at distance r from the centre line, and P lies $r\sin\theta$ closer to it than O does. So

$$a = r - r\sin\theta = r(1-\sin\theta) \ge r\left(1-\tfrac{4}{5}\right) = \tfrac{1}{5}r,$$

which gives $r \le 5a$.

Checks

- At the extreme case $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$, take $W = 1$ and $k = 1$. The plank equation gives $F = \frac{1}{4}$, then $R = \frac{1}{2}$ and $N = \frac{3}{2}$. So $G/R = \frac{1}{2}$ exactly (the plank is on the point of slipping) while $F/N = \frac{1}{6}$, comfortably below $\frac{1}{2}$, as part (ii) predicts.
- All four equilibrium equations (moments, and resolving for both bodies) were confirmed by computer for 10^5 random choices of W, k and θ. So was the claim that $G \le \frac{1}{2}R$ holds exactly when $\sin\theta \le \frac{4}{5}$, and that $F < \frac{1}{2}N$ always holds then.
- Limiting sense: a wide plank ($a \to r$) sits near the tops, so $\theta \to 0$ and almost no friction is needed. A narrow plank wedges deep between the cylinders, and eventually it slides down between them.

Common slips

- Leaving out the moment equation and so not realising that $G = F$. Without it there are too many unknowns.
- Getting the direction of the plank's friction wrong: the cylinders tend to roll apart, so on the cylinder it acts downwards and inwards.
- Writing $a = r\sin\theta$ rather than $a = r(1-\sin\theta)$: a is measured from the centre line, not from the cylinder's centre.

Beyond STEP

This is a small statically determinate system: three bodies, enough equations to fix every force, and inequalities from friction that mark out the possible configurations. In a first-year course the same ideas reappear as the conditions for rigid-body equilibrium (zero total force and zero total moment), and later in the study of wedging and jamming. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

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